

Dear Industrious Engineering team,

I am writing to express my interest in the electrical engineering graduate position. As an enthusiastic final year Mechatronics Engineering and Applied Physics student, I believe my background and passions align well with the role. Specifically, I have a strong interest in the hardware engineering and hope to contribute to energy sustainability projects. With my experience in many electronics projects, and my strong drive to gain new skills, I believe I would be a valuable new member of your team.

A commitment to ethical conduct and the highest standards of professional accountability

A rigorous set of safety rules and requirements are in place for the competition that UTS Motorsports Electric participates in. One of these dictates the section and thickness requirements of the roll cage of the car. When the car construction was almost complete, a chance inspection found that the steel tube thickness we were using was too small. Although the cars are inspected as part of the competition, this error could easily be overlooked and probably would not be flagged. As part of the team leadership, I had to help decide how to proceed, given that rectifying the mistake would be costly and time consuming. Of course, the tube was replaced, which was uncomfortable because it meant that the budget for some other areas had to be reduced. The purpose of the rules is to ensure the design is as safe as possible, and this issue prompted our decision to recheck all rules and ensure factors of safety were present on all suspension components.

Demonstrated ability to effectively communicate both with other engineers and with stakeholders from different fields

During the initial design of puzzle automation platforms at Labyrinth Escape Rooms, I had to work with stakeholders who designed the puzzles, casual workers who interacted with customers, and other engineers. It was important to communicate technical limitations effectively without jargon, when defining the scope of the project. I learned that rather than ‘dumbing down’ technical concepts, it was valuable to patiently explain them, as this fostered a deeper understanding between stakeholders. Another aspect of this job was to create and teach workers to use interfaces. Description, demonstration, and documentation were the guiding principles I tried to follow to make things clear for users.

The ability to engage with a creative, innovative and proactive environment

At Labyrinth Escape Rooms I worked as part of a small team of engineers to develop a web interface for automation of many small, embedded platforms. The fast-paced development of multiple interfaces in several different languages was a very rewarding experience which allowed me to get acquainted with languages including C, JS, and Python. The environment encouraged a team of equals to contribute ideas, discuss freely and try rather than hypothesize. I was able to offer my own ideas and feel heard and trusted while also practicing the ability to put aside my own opinions and be obedient when required.

Demonstrated ability to use and manage information

At UTS Rapido my first task was the development of a dissolved oxygen sensor. It was important to clearly lay out the design process which consisted of several electrical sections each with different calculations and design decisions. Rather than just including calculations in an excel sheet, I tried to create documentation which had a logical flow and was clearly described. Each section included the goal of the module, relevant information sources and calculations. Finally, a summary of how the calculations affected design decisions was included for each section. This style of documentation ensured that handovers to other engineers was smooth easy and was well received by coworkers.

The ability to manage your own performance in a professional environment

As part of automating puzzles at Labyrinth Escape Rooms, developing a work plan and breaking tasks into milestones was a key step to ensure that work was completed on time and correctly. To manage

working on 10+ separate embedded systems and interfaces simultaneously I used the GitHub Project Planning tool. This allowed me to collaboratively track many modules at the same time and stay on schedule. Inevitably, things did not always go as planned and deadlines occasionally became unattainable. When this happened, I would discuss with my team and supervisor to understand why something is taking longer than expected and how it can be solved. When faced with these blockages, an important step was to consider the approach and whether it was time to try a different direction. Being thorough and disciplined is necessary, but knowing when to be flexible was also important. Using a scheduling tool to set limits helped in knowing how to apply myself to these tasks.

A demonstrated ability to work as part of a team and show leadership when required

At UTS Motorsports Electric I worked as part of a large student team to design and build an electric race car. As the vehicle dynamics section leader, I gained experience with team leadership responsibilities, working to deadlines and communicating with colleagues, supervisors, and suppliers to solve problems. By creating an in-depth schedule and arranging weekly meetings, we were able to successfully complete the year-long project including design, build and test phases. I think that creating an environment of discussion and support rather than focusing on delegation helped to create a more creative and productive team.

I believe that I have strong fundamentals and research skills and bring considerable experience with projects to be an asset. Most importantly I am deeply passionate about developing new skills and I believe that Industrus offers a valuable opportunity to work with enthusiastic and knowledgeable people. I am eager to hear back from you to arrange an interview.

Sincerely,

Jesse McNamara